6 **(a)** A student claims that "Light is only a wave and not a particle."

Discuss whether this statement is correct, providing experimental evidence.

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.....[3]

**(b)** A cool, low-pressure gas of an element X is illuminated by a beam of monochromatic violet light of frequency $8.21 \times 10^{14}$ Hz.

The simplified energy level diagram for an atom of element X is shown in Fig. 6.1 below.

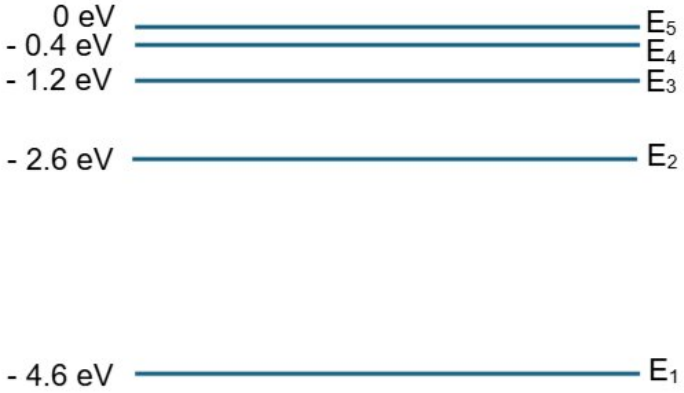



**Fig. 6.1**

**(i)** Calculate the energy of a photon of the incident violet light.

energy of a photon = ..... eV [2]

**(ii)** Draw, on Fig. 6.1, the transition corresponding to the absorption of the incident violet photon. [1]

**(iii)** Determine the number of different emission lines that may be observed as a result of the absorption of many of such violet photons.

number of emission lines = ..... [1]

**(iv)** Explain why the intensity of the violet light decreases after passing through the gas.

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.....[2]

[Total: 9]